

Homework (Carolina Reaper)

- Q1.** A music artist is creating a rhythm pattern with $3^3 + 5^3$ beats. What is the factored form of this expression?
(A) $8(3 + 5)$
(B) $8(3 - 5)$
(C) $(3 + 5)(9 - 15 + 25)$
(D) $(3 - 5)(9 + 15 + 25)$
(E) $(3 + 5)(9 + 15 - 25)$
- Q2.** A painter is designing a canvas with dimensions $x^3 - 4^3$. What is the factored form of this expression?
(A) $x(3 - 4)$
(B) $x(3 + 4)$
(C) $4(x - 4)(x^2 + 4x + 16)$
(D) $4(x + 4)(x^2 - 4x + 16)$
(E) $4(x - 4)(x^2 + 4x + 16)$
- Q3.** A sound engineer is mixing frequencies at $2^3 + 6^3$ Hz. What is the factored form of this expression?
(A) $8(2 - 6)$
(B) $8(2 + 6)$
(C) $(2 + 6)(4 - 12 + 36)$
(D) $(2 - 6)(4 + 12 + 36)$
(E) $(2 + 6)(4 + 12 - 36)$
- Q4.** A music producer is creating a beat with $x^3 - 9^3$ notes. What is the factored form of this expression?
(A) $9(x - 9)(x^2 + 9x + 81)$
(B) $9(x + 9)(x^2 - 9x + 81)$
(C) $9(x - 9)(x^2 - 9x + 81)$
(D) $9(x + 9)(x^2 + 9x + 81)$
(E) $9(x + 9)(x^2 - 9x - 81)$
- Q5.** A artist is designing a sculpture with $5^3 + 2^3$ pieces. What is the factored form of this expression?
(A) $7(5 + 2)$
(B) $7(5 - 2)$
(C) $(5 + 2)(25 - 10 + 4)$
(D) $(5 - 2)(25 + 10 + 4)$
(E) $(5 + 2)(25 + 10 - 4)$
- Q6.** A composer is writing a piece with $x^3 - 8^3$ notes. What is the factored form of this expression?
(A) $8(x - 8)(x^2 + 8x + 64)$
(B) $8(x + 8)(x^2 - 8x + 64)$
(C) $8(x - 8)(x^2 - 8x + 64)$
(D) $8(x + 8)(x^2 + 8x + 64)$
(E) $8(x + 8)(x^2 - 8x - 64)$
- Q7.** A painter is creating a mural with $3^3 + 7^3$ colors. What is the factored form of this expression?
(A) $10(3 - 7)$
(B) $10(3 + 7)$
(C) $(3 + 7)(9 - 21 + 49)$
(D) $(3 - 7)(9 + 21 + 49)$
(E) $(3 + 7)(9 + 21 - 49)$
- Q8.** A music artist is designing a rhythm pattern with $x^3 + 3^3$ beats. What is the factored form of this expression?
(A) $3(x + 3)(x^2 - 3x + 9)$
(B) $3(x - 3)(x^2 + 3x + 9)$
(C) $3(x + 3)(x^2 + 3x + 9)$
(D) $3(x - 3)(x^2 - 3x + 9)$
(E) $3(x + 3)(x^2 - 3x - 9)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $x^3 - 2^3$ and explain the process used to factor it.
- Q10.** Factor the expression $4^3 + x^3$ and provide an example of a real-world application of this factored form in the arts or music.

Chef's Tips

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- Tip 1: Ensure students understand the formula for factoring the sum and difference of cubes.
- Tip 2: Encourage students to use real-world examples to illustrate their understanding of the factored forms.
- Tip 3: Emphasize the importance of checking their work by multiplying the factored forms to ensure they equal the original expression.